

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class : S.Y.I.T.	Batch : 1
Date of Assignment : 12/08/2026	Date/Time of Submission : 12/08/2026

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

CODE :

```
import pandas as pd

data = {
    "name": ["Akku", "Nikku", "Sahil"],
    "age": [50, 40, 30],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)
print(df)
print("Abhishek S025")
```

OUTPUT :

```
   name  age  qualified
0  Akku   50      True
1  Nikku  40     False
2  Sahil  30     False
Abhishek_S025
```

5b. Write a python code to show statistical information on given data set.

CODE :

```
import statistics
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
print(statistics.stdev([1, 30, 50, 100]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print("Abhishek_S025")
```

OUTPUT :

```
7
1.05
red
41.67633221226007
0.47966666666666663
Abhishek_S025
```

5c. Write a python code to create pandas series from dictionaries.

CODE :

```
import pandas as pd

dictionary = {'A': 10, 'B': 20, 'C': 30}

series = pd.Series(dictionary)

print(series)
print("Abhishek_S025")
```

OUTPUT :

```
A      10
B      20
C      30
dtype: int64
Abhishek_S025
```

CODE :

```
import pandas as pd

dictionary = {'A': 50, 'B': 10, 'C': 80}

series = pd.Series(dictionary, index=['B', 'C', 'A'])

print(series)
print("Abhishek_S025")
```

OUTPUT :

```
B      10
C      80
A      50
dtype: int64
Abhishek_S025
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

CODE :

```
import pandas as pd

dict = {'name':["Akku", "Sahil", "Nikku", "Krish"],
        'degree': ["MBA", "BCA", "B.Tech", "MBA"],
        'score':[90, 100, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
print("Abhishek_S025")
```

OUTPUT :

	name	degree	score
True	Akku	MBA	90
False	Sahil	BCA	100
True	Nikku	B.Tech	80
False	Krish	MBA	98

Abhishek_S025